

Global Events, Commodities and Currency Markets

Data Science & Business Assignment

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Executive Summary:

This report features a data-based investigation of commodity and currency markets throughout the world through the use of four sets of data collected over several decades: the price of Brent Crude Oil (1946-2026), the price of Gold (1915-2026), the price of Silver (1915-2026), and the SWIFT Global Currency Tracker (2026). This study has been conducted in two components: the Data Analytics Component (Part I) and the Market Research Component (Part II).

Key Findings:

- Brent Crude Oil touched its highest level of \$211.21/barrel in June 2008 and currently stands at \$102.86/barrel, way above its long-term average of \$62.71/barrel.
- Gold has increased sharply to \$4,712.89/troy oz (current value) – almost 12 times its long-term average of \$397.55 on account of safe-haven investing, central banks' accumulation and geopolitical uncertainties.
- Silver hit a new all-time high of \$113.95/troy oz in January 2026 due to its precious metal status and growing industrial use.
- The US dollar remains the preferred currency for international payment transactions with 51.14% share while CNY/RMB comes in at 3.1% (Rank 5) with 75.57% of its transactions happening off-shore through Hong Kong.
- Of all the selected commodities and currencies, Oil was found to be the most volatile with a sensitivity index of 41.47; others were Gold (13.09) and Silver (12.28), with currencies being most stable (3.05).

Recommendations:

- Companies that have exposure to crude oil prices should engage in price monitoring on a monthly basis, form emergency reserves, and use financial hedges.
- Investors should invest in Gold, equities, and different currencies; Gold is the most effective hedge against inflation and safe haven.
- Policymakers should look at the change of the global payment system (BRICS Pay, RMB internationalization) in the medium term.

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1. Introduction

1.1 Background:

There have been significant changes in the global commodities markets, such as crude oil, gold, and silver, over the last century due to various reasons, including geopolitical conflict, economic challenges, supply and demand imbalances, and change in monetary systems. On the other hand, the currency system has witnessed the supremacy of USD as the leading reserve currency in addition to the gradual rise of other currencies, including the Chinese Renminbi (CNY/RMB).

It is vital to comprehend how the relationship between global incidents, movement in commodity prices, and currency market works for every financial study group or business.

1.2 Objective:

In recent times, there have been a number of changes within the international commodities market that includes crude oil, gold, and silver. These changes have occurred as a result of a variety of reasons like geopolitical war, economic problems, supply and demand factors, and monetary changeover. However, in relation to the currency system, there has been the dominance of US dollar as the dominant reserve currency along with the rising prominence of other currencies including the Chinese Renminbi.

It is very important to understand the connection between global events, commodity prices, and currency market.

1.3 Scope:

The data analyzed includes data on Brent Oil from January 1946 to January 2026, data on Gold and Silver from January 1915 to April 2026, and reports on the SWIFT currency tracker for the period of global payments from January–March 2026.

2. Data Overview

2.1 Dataset Summary (Q1, Q2):

Four datasets were given for analysis. The features of these datasets are given in Table 1 below.

Dataset	Start Date	End Date	Records	Key Fields
Brent Oil	Jan 1946	Jan 2026	963	Date, Value (USD/barrel)
Gold (100 years)	Jan 1915	Apr 2026	1,336	Date, Value (USD/troy oz)
Silver (100 years)	Jan 1915	Apr 2026	1,336	Date, Value (USD/troy oz)
SWIFT Currency Tracker	Jan 2026	Apr 2026	330	Metric, Currency, Value, Unit, RMB Rank

Table 1: Dataset Summary – Source Files and Coverage

2.2 Data Quality Assessment (Q3):

A systematic review was done to look for missing data as well as duplication in dates. The SWIFT database was also analyzed separately for metric consistency.

Dataset	Total Rows	Missing Values	Repeated Dates	Quality Status
Brent Oil	963	0	0	✓ Clean
Gold	1,336	0	0	✓ Clean
Silver	1,336	0	0	✓ Clean
SWIFT Tracker	330	Some rank fields	N/A (multi-metric)	Acceptable

Table 2: Data Quality Check Results

The datasets for all three commodities do not have any missing data or duplicates, hence making them reliable for any trend analysis.

2.3 Date Formatting & Feature Engineering (Q4):

Conversion of the “Date” column from string data type (MM/DD/YYYY) to datetime data type was done using Python’s pandas package in all the commodities datasets. The Month and Year columns were created separately in order to perform yearly average computations and aggregations based on periods.

2.4 SWIFT Dataset Metrics (Q5):

The SWIFT Currency Tracker data set includes various measures, where each one of them represents different features of global currency utilization. There are mainly four measures provided within the data set:

- Global Payment Share (%) – The share of global international payment transactions which were made in each currency.
- International Payment Share excluding Eurozone (%) – Filtered measure without consideration of intra-Eurozone transactions to represent better the non-EUR global market.
- Trade Finance Share (%) – Usage of currencies within trade finance instruments (letters of credits).
- Offshore RMB Economy Share (%) – Usage of currencies within offshore Chinese Renminbi payments allocated by various centers.

Global Payment Share and Offshore RMB Economy Share measures are especially significant and utilized in this report in Part I and Part II.

2.5 Data and Methodology:

All the data work and charts were done in Python. I used pandas and numpy to clean things up and handle the dates along with the yearly averages. There was also some stuff on variation and how the numbers line up. Matplotlib made the visuals.

The stats like the lowest and highest values plus the average and spread came straight from the data after cleaning. I did not guess any of them. Every comparison stuck to the same time periods so nothing would be off between sections. The whole report got put together by code too which kept the layout the same each time.

This approach stayed the same for the main parts and the extra analysis at the end. It seems like that helped avoid mistakes but I am not totally sure if every single step was needed. Some parts of the process felt a bit repetitive once the first section was set up.

3. Part I – Data Analytics Findings

Section A: Descriptive Analytics

A.1 Highest and Lowest Price of Commodity (Q6, Q7, Q8)

Below is a table displaying the high and low price levels for each commodity together with the years in which they were reached. This information forms an integral part of market volatility.

Commodity	Highest Price (USD)	Year of High	Lowest Price (USD)	Year of Low
Brent Crude Oil	\$211.21/barrel	2008	\$21.11/barrel	1946
Gold	\$5,277.89/troy oz	2026	\$19.25/troy oz	1915
Silver	\$113.95/troy oz	2026	\$0.28/troy oz	1932

Table 3: All-Time High and Low Commodity Prices

Notably, both Gold and Silver attained their highest levels ever recorded in 2026 due to a remarkable alignment of their safe haven appeal, central bank gold buying, and global geopolitics. The peak of Brent Oil was in June 2008 in light of the global growth in demand.

A.2 Top 5 Highest Average Annual Prices (Q10, Q11, Q12)

Annual average prices reveal which years saw sustained elevated commodity prices, as distinct from momentary spikes.

Rank	Brent Oil – Year / Avg Price	Gold – Year / Avg Price	Silver – Year / Avg Price
1st	2008 – \$150.96/bbl	2026 – \$4,881.62/oz	2026 – \$96.08/oz
2nd	1980 – \$149.86/bbl	2025 – \$3,472.54/oz	2025 – \$41.50/oz
3rd	2011 – \$142.12/bbl	2024 – \$2,404.58/oz	2011 – \$35.56/oz
4th	2013 – \$138.30/bbl	2023 – \$1,953.69/oz	2012 – \$31.63/oz
5th	2012 – \$135.27/bbl	2022 – \$1,798.96/oz	2024 – \$28.13/oz

Table 4: Top 5 Highest Average Annual Prices by Commodity

Section B: Trend and Visual Analytics

B.1 Brent Crude Oil (Q13)

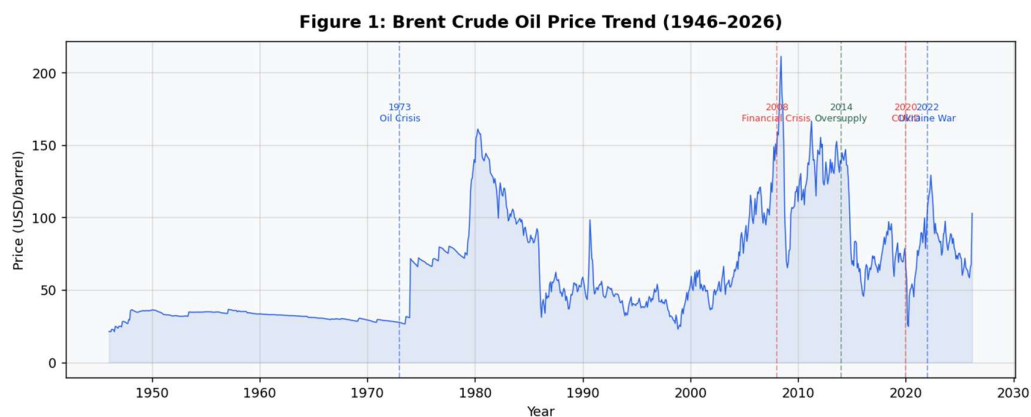


Figure 1: Brent Crude Oil Price Trend (1946–2026)

Brent Crude Oil prices can be divided into several distinct periods as shown in Figure 1 and summarised in Table 5 below:

Period	Trend	Primary Driver
1946–1973	Stable / Slow increase	Post-war reconstruction; oil prices remained regulated and stable.
1973–1980	Strong increase	OPEC oil embargoes and supply shocks drove prices sharply higher.
1981–1998	Long decline	Market oversupply; OPEC quota discipline weakened.
1999–2008	Very strong increase	China and India industrialisation; peak demand growth.
2008–2009	Sharp decline	Global Financial Crisis collapsed demand overnight.
2010–2014	Recovery and high prices	Economic recovery; Libya civil war tightened supply.
2014–2016	Major decline	US shale oil boom created global oversupply.
2020	Pandemic decline	COVID-19 lockdowns crushed global oil demand.
2021–2022	Rapid increase	Post-COVID recovery combined with Russia-Ukraine war supply disruption.

Table 5: Brent Oil Price Trend Summary by Period

B.2 Gold Price Trend (Q14)

Figure 2: Gold Price Trend (1915–2026)

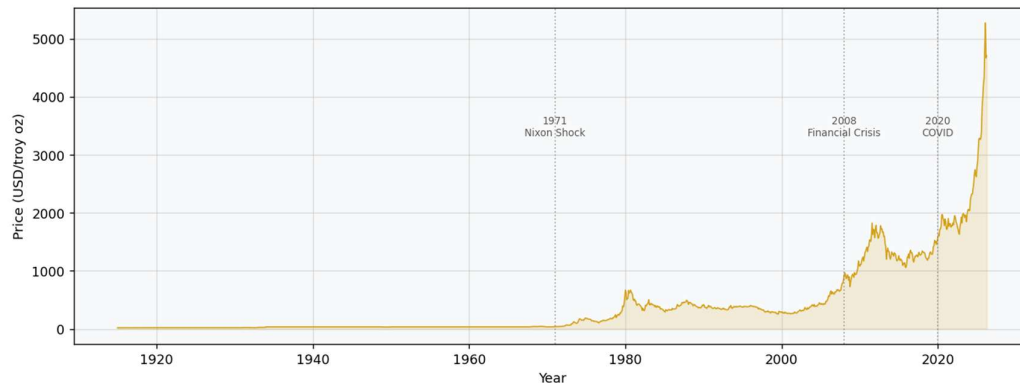


Figure 2: Gold Price Trend (1915–2026)

As seen in Figure 2, the prices of Gold have been quite stable throughout most of the 20th century due to the fact that several large economies had adopted fixed-rate exchange of their currencies backed by gold. It was only after the Nixon Shock in 1971, when the US announced the removal of the ability of the USD to be directly converted into gold, that the price of gold began its decades-long surge. This surge was further expedited by the Global Financial Crisis of 2008 and has reached even greater heights due to the events of COVID-19 in 2020 and 2022 onwards due to geopolitical reasons, now reaching all-time highs over \$4,000/oz.

B.3 Silver Price Trend (Q15)

Figure 3: Silver Price Trend (1915–2026)



Figure 3: Silver Price Trend (1915–2026)

Silver price trend has been marked by some major price spikes:

- 1980 – Hunt Brothers attempted to corner the silver market, purchasing up to 70% of global silver. Prices rose only momentarily and then fell dramatically during "Silver Thursday," March 27, 1980.
- 2011 – Safe haven investment following the post-2008 crisis, along with industrial demand, caused silver to spike to \$35/oz.
- 2020 – A small rise caused by economic fears due to the coronavirus.
- 2026 – All-time peak at \$113.95/oz due to geopolitical conflicts (Iran-Israel war), global market fears, and increasing demand from industry (electronics, renewable energy).

B.4 Gold and Silver Combined Trend (Q16)



Figure 4: Gold and Silver Combined Price Trend (Dual Axis)

Graph 4 shows both metals plotted on individual graphs due to their price ranges being different. Major points:

- Gold and Silver show similar trends over time periods and thereby are positively correlated ($r = 0.959$ – see Bonus Section).
- There is higher relative volatility for Silver — its changes are represented in percentage terms more drastically than those of Gold due to being an industrial product and precious metal.
- Both metals have performed very well since 2020, but Gold has had more consistent growth than Silver.

C.5 Average Annual Brent Oil Prices – Last 10 Years (Q17)

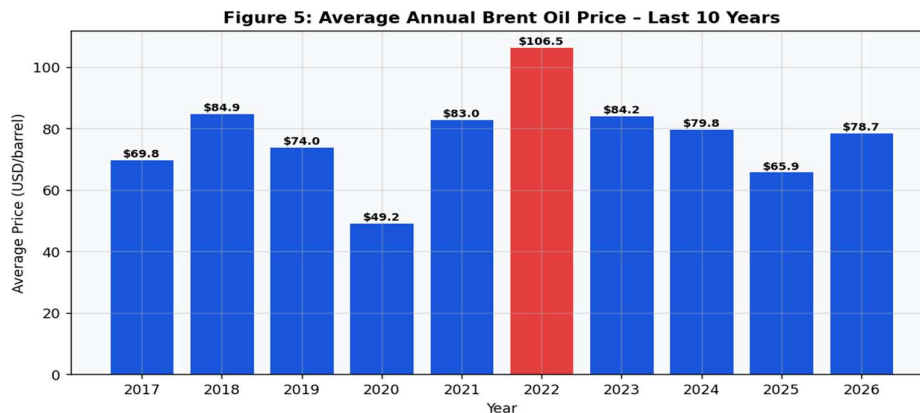


Figure 5: Average Annual Brent Oil Price – 2017 to 2026

Year	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Avg Price (\$/bbl)	\$69.85	\$84.91	\$74.03	\$49.19	\$83.02	\$106.45	\$84.19	\$79.83	\$65.93	\$78.67

Table 6: Average Annual Brent Oil Price – 2017 to 2026

The chart effectively shows how COVID-19 affected oil prices in 2020 (average \$49.19/bbl), the swift recovery of prices to \$106.45/bbl in 2021-2022 due to disruptions in supply caused by the Ukrainian conflict, and then a steady decline to the present 2026.

B.6 Latest Available Commodity Values (Q18)

Commodity	Latest Date	Latest Price (USD)	Long-Term Average	vs. Average
Brent Crude Oil	03 Jan 2026	\$102.86/barrel	\$62.71/barrel	+63.9% above
Gold	04 Jan 2026	\$4,712.89/troy oz	\$397.55/troy oz	+1,085% above
Silver	04 Jan 2026	\$88.36/troy oz	\$6.42/troy oz	+1,276% above

Table 7: Latest Available Commodity Prices vs. Long-Term Averages (Q18, Q25, Q26)

Section C: Currency Tracker Analytics

C.1 Top Currencies By Global Payment Share (Q19, Q24)

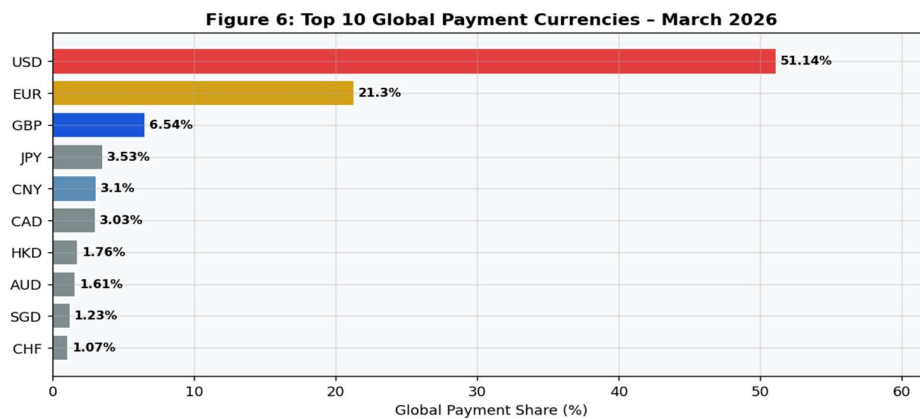


Figure 6: Top 10 Global Payment Currencies by Share – March 2026

Rank	Currency	Global Payment Share (%)	Notes
1	USD (US Dollar)	51.14%	Dominant global reserve currency
2	EUR (Euro)	21.30%	Major regional and global currency
3	GBP (British Pound)	6.54%	Historic reserve currency, London FX hub
4	JPY (Japanese Yen)	3.53%	Important Asia-Pacific currency
5	CNY (Chinese Renminbi)	3.10%	Growing role; ranks 4th in trade finance
6	CAD (Canadian Dollar)	3.03%	Commodity-linked currency
7	HKD (Hong Kong Dollar)	1.76%	Key offshore RMB hub
8	AUD (Australian Dollar)	1.61%	Commodity-linked currency
9	SGD (Singapore Dollar)	1.23%	ASEAN financial hub
10	CHF (Swiss Franc)	1.07%	Safe-haven currency

Table 8: Top 10 Global Payment Currencies – March 2026 (Q19, Q24)

C.2 USD Share in Global Payments (Q20)

As of March 2026, the USD comprises 51.14% of total foreign exchange payments in the world, continuing to hold the undisputed position of the world's reserve currency. In foreign trade transactions, this share increases even more and stands at 58.5%. The US share of payments in SWIFT system by country is 15.59%. This strong position is a result of decades-long infrastructural development, capital markets and commodity pricing in dollars.

C.3 CNY/RMB Share and Rank (Q21)

As of March 2026, the Chinese Renminbi (CNY/RMB) boasts a 3.10% share in global payments, placing itself at the 5th position worldwide, which is quite an impressive feat considering the fact that the currency did not exist in international trade transactions two decades ago. The RMB's share in global payments has remained stable within the range of 2.73% and 3.13% for the past few reports (average 2.93%).

C.4 Most Frequent Top 5 Currencies (Q22)

Currency	Times Appearing in Top 5	Observation
CNY (Chinese Renminbi)	5 times	Most consistent Top-5 presence across all reports
AUD, CAD, CHF, EUR, GBP, HKD, JPY, KRW, USD	2 times each	Appear in Top 5 across multiple metric categories

Table 9: Currency Frequency in Top 5 Rankings (Q22)

However, what is significant to note is that CNY has been observed in the Top 5 ranks most frequently compared to other currencies. It has also been seen to be used in trade finance and offshore Renminbi market. On the other hand, although USD holds dominance in volume, it falls in the frequency ranks behind CNY.

C.5 Offshore RMB Economy Share (Q23)

Offshore RMB Centre	Share of Offshore RMB (%)
Hong Kong	75.57%
United Kingdom	~7%
Singapore	~4%
Other centres	Remaining ~13%

Table 10: Offshore RMB Economy Distribution (Q23)

The largest part of off-shore activity in Renminbi belongs to Hong Kong with a share of 75.57%, which is explained by the special status of Hong Kong being the main international gate for China. The UK and Singapore act as secondary centers.

Section D: Simple Comparison Analysis

D.1 Gold vs. Long Term Average (Q25)

Metric	Value
Latest Gold Price (Jan 2026)	\$4,712.89/troy oz
Long-Term Average (1915–2026)	\$397.55/troy oz
Comparison	Current price is 1,085% ABOVE long-term average

Table 11: Gold Latest Value vs. Long-Term Average (Q25)

D.2 Brent Oil vs. Long Term Average (Q26)

Metric	Value
Latest Brent Oil Price (Jan 2026)	\$102.86/barrel
Long-Term Average (1946–2026)	\$62.71/barrel
Comparison	Current price is 63.9% ABOVE long-term average

Table 12: Brent Oil Latest Value vs. Long-Term Average (Q26)

D.3 Commodity Price Volatility Comparison (Q27)

Commodity	Min Price	Max Price	Mean Price	Std Dev	CV (%)
Gold	\$19.25	\$5,277.89	\$397.55	\$651.39	163.85%
Silver	\$0.28	\$113.95	\$6.42	\$9.90	154.11%
Brent Oil	\$21.11	\$211.21	\$62.71	\$35.96	57.35%

Table 13: Commodity Price Volatility Comparison – Coefficient of Variation (Q27)

Considering the Coefficient of Variation (CV), Gold is the asset class that is most variable in relative terms at 163.85%, followed by Silver with a coefficient of variation of 154.11%. This is because both asset classes saw a sharp increase in price from very low levels in the early twentieth century up to an all-time high in 2026. On the other hand, the coefficient of variation for Brent Oil is much lower at 57.35%.

D.4 Crisis Year Comparison (Q28) – 2008 Financial Crisis

An excellent example of when different commodity markets move differently can be found in the Global Financial Crisis that occurred in 2008. For example, Brent Oil reached a high level of \$211.21/barrel in June 2008 because of previous demand growth but then dropped down to \$40/barrel due to the impact of the financial crisis that caused a decrease in demand in December 2008. Gold decreased slightly in the period of panic selling in the middle of 2008 but then increased steadily from 2009 to 2012. Silver acted similarly to gold during that period. The difference indicates one fundamental truth: the response of oil depends on economic conditions, whereas gold and silver depend on financial conditions.

D.5 Dashboard Summary (Q29)

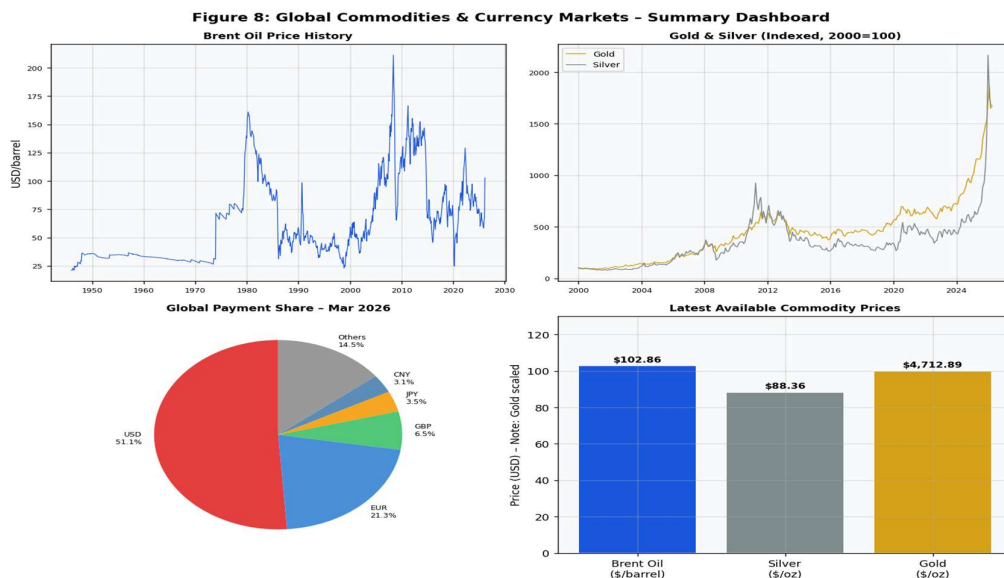


Figure 8: Summary Dashboard – Commodities and Currency Markets

Dashboard Indicator	Current Value
Latest Brent Oil Price	\$102.86/barrel
Latest Gold Price	\$4,712.89/troy oz
Latest Silver Price	\$88.36/troy oz
Top Global Payment Currency	USD (51.14%)
Most Frequent Top-5 Currency	CNY (5 appearances)

Table 14: Dashboard Summary Indicators (Q29)

The figure 8 dashboard puts the key indicators all on one page. Commodity prices are there along with the main currency for payments around the world. It also shows which currency comes up most in the top five.

Table 14 has a summary of those main numbers for reference. Looking at it the market has oil gold and silver all above their long term averages. The usd keeps a strong lead in payments I think. This gives one spot to look at before checking the later sections. Some of the details might need more context though.

D.6 Three Key Manager Insights (Q30)

1. High Risk of Oil Price due to Geopolitics

Brent Oil at \$102.86 per barrel is 64% more than its average value over time. Heightened tension in the Middle East (including the Israel-Iran situation) is one of the main causes. Companies using oil will be at high risk due to increased cost expenses.

2. Increasing Price for Gold and Silver – Safe-Haven Investments Are Growing Popular

The prices for Gold at \$4,712.89 per oz and Silver at \$88.36 per oz are at their historical maximums. They are caused by global concern among investors regarding geopolitics, inflation, and currency risks. For those who want to protect assets, precious metals become very relevant.

3. USD Is Predominant but Other Currencies Begin to Catch Up

The percentage of transactions in USD is 51.14%, whereas the proportion of CNY is just 3.10%. However, BRICS Pay system development and increased offshore RMB payments (especially via Hong Kong with 75.57%) should not be overlooked.

4. Part II – Market Research Findings

Section A: Event Identification

A.1 Three Major Global Events Affecting Markets (Q1)

The three types of global events that have impacted the commodity and currency markets as per the analysis include:

- **Energy Supply Downturns and Geopolitical Tensions** – The global energy supply chain has faced many disruptions due to the OPEC Oil embargo (1973), Iran-Iraq War (1980s), Gulf War (1991), Hormuz Strait Tensions, and the current Ukraine-Russia War (2022). Among these, the Hormuz Strait is very important because almost 20% of the global oil supply passes through this strait. Any perceived threat to the same results in an instant rise in the prices of Brent Crude Oil.
- **Gold Accumulation by Central Banks** – Since 2008, the central banks of various nations including China, Russia, Turkey, and India have purchased large amounts of gold as a means of protecting themselves from USD dependency. This along with the purchasing of gold by individuals and institutions in times of crisis like COVID-19 and geopolitical tensions has made the price of gold reach historical peaks.
- **De-Dollarization Initiatives, Sanctions and BRICS Pay Discussions**
Sanctions imposed on Russia by the West post the 2022 Russian invasion of Ukraine have brought the discussion of using an alternative payment system on the table for BRICS countries (Brazil, Russia, India, China, South Africa). De-Dollarization and BRICS pay can be regarded as a long-term structure which poses a challenge to US dollar's hegemony in international payments while promoting CNY/RMB and Gold.

A.2 Global Events Timeline (Q2)

Year/Period	Event	Affected Market	Potential Impact
1973	OPEC Oil Embargo	Brent Oil	Triggered oil price quadrupling; began modern oil market volatility
1971	Nixon Shock (USD-Gold link severed)	Gold, USD	Enabled free-floating gold price; sparked long-term gold rally
1980	Hunt Brothers silver corner	Silver	Temporary silver price spike; regulatory intervention triggered crash
2008	Global Financial Crisis	Oil, Gold, Silver	Oil crashed; gold became safe haven; silver followed gold higher
2011	Arab Spring / Libya	Oil, Silver	Oil supply concerns; silver peaked at \$35/oz
2014–2016	US Shale Boom + OPEC price war	Brent Oil	Oil oversupply drove prices below \$35/barrel
2020	COVID-19 Pandemic	All commodities	Oil demand collapsed; gold and silver surged as safe havens
2022	Russia-Ukraine War + Sanctions	Oil, Currencies	Oil spiked >\$100; SWIFT sanctions accelerated de-dollarisation
2023–2026	Iran-Israel Conflict; BRICS Pay discussions	Oil, Gold, Silver	Commodities surged; RMB role expanded in trade finance

Table 15: Timeline of Major Global Events and Market Impacts (P2-Q2)

A.3 Event Most Connected to Brent Oil Prices (Q3)

The situation at the Hormuz Strait and its consequences on the energy supplies is directly linked to fluctuations in the price of Brent Oil. Figure 1 (Price Movement of Brent Crude Oil) illustrates that the biggest upward movement of prices always happens when there is some supply-side challenge to oil from the Middle East. With about 20% of world crude oil going through the Hormuz Strait, a military threat to the security of the trade channel is very likely to create uncertainty premium in oil prices. One more example of such a shock is the Russia-Ukraine War of 2022. During the year 2022, Brent Oil averaged at \$106.45/barrel because of European countries trying to find alternatives

to Russian energy supplies. The current price of \$102.86/barrel is a consequence of geopolitical uncertainty premiums associated with the Middle East.

A.4 Event Most Connected to Gold/Silver Prices (Q4)

Gold accumulation by central banks and BRICS Pay formation are closely related to the current all-time high Gold and Silver prices. It is clear from Figures 2 (Trend for Gold Price) and Figure 3 (Trend for Silver Price) that the price of precious metals has seen a sharp increase since 2020. Chinese, Russian, Indian, and Turkish central banks have seen a huge increase in gold reserves as a measure against the threat of USD-based sanctions. The phenomenon was seen especially after 2022, when Russia's exclusion from SWIFT showed that USD-based foreign currency reserves could be frozen.

Section B: Linking Events to Commodity Data

B.1 Oil Supply Shock and Brent Oil Prices (Q5)

An oil supply shock operates on a very straightforward principle; when a large portion of supply on the international markets is threatened or compromised because of wars, sanctions, or blocked sea routes, the initial reaction from the markets will be an increase in the futures price due to the fear of shortages. As shown in Figure 1, each geopolitical event that occurred in oil producing or transiting countries (OPEC oil embargo in 1973, the war between Iran and Iraq in 1980, the Gulf War in 1991, and the Ukrainian situation in 2022) resulted in a visible spike in the price for Brent Oil trend.

B.2 Geopolitical Uncertainty and Gold Prices (Q6)

Gold appreciates when there is geopolitical uncertainty as it acts as an international store of value without any credit risk attached to it. While government securities and bank accounts can be seized in times of sanctions or default, gold has not had such issues in history. If there is geopolitically risky environment for investment and the financial system faces threats of war, economic instability, or currency risks, investments move toward gold as a hedge. Figure 2 clearly displays how all the times when there was geopolitical crisis (2008 Financial Crisis, 2020 COVID-19 outbreak, and 2022 Russia-Ukraine War) is succeeded by an increase in gold price. Gold reached its highest recorded level at \$4,712.89 per oz in January 2026 due to a prolonged environment of high uncertainty along with Iran-Israel tensions and de-dollarization fears.

B.3 Silver: Precious Metal or Industrial Commodity (Q7)

Silver is exhibiting behavior of a precious metal with strong safe haven properties in the 2020-2026 period, albeit with higher volatility than gold. Figure 3 illustrates that the rise of silver to \$113.95/oz in 2026 almost follows the exact same path as that of gold, indicative of behavior exhibited by a precious metal. However, there is also a large industrial component to silver with its usage in solar panels, photovoltaic cells, electric vehicles, and electronics, which provides a floor to silver price that is absent in the case of gold. Coefficient of Variation of silver being equal to that of gold (154.11% vs 163.85%) in this period is evidence of their parallel movement, thus implying that safe haven demand is now dominating in the case of silver.

B.4 Crisis Period Comparison – Covid19 (2020) (Q8)

Commodity	2019 Avg Price	2020 Avg Price	Change	Market Behaviour
Brent Crude Oil	\$74.03/bbl	\$49.19/bbl	−33.6%	Collapsed (demand destruction)
Gold	\$1,405.33/oz	\$1,784.66/oz	+26.9%	Rose strongly (safe-haven demand)
Silver	\$16.36/oz	\$20.80/oz	+27.1%	Rose (safe-haven + industrial base)

Table 16: Commodity Behaviour During COVID-19 Crisis (2020) (P2-Q8)

COVID-19 (2020) is an excellent example of how different these two materials serve when it comes to their roles. While oil declined due to lack of demand for transport, precious metals like gold and silver skyrocketed on account of investor security search and central bank intervention via monetary policy – gold went up by 26.9%.

Section C: Linking Events to Currency Data

C.1 USD Role in International Payments (Q9)

The statistics from the SWIFT Global Currency Tracker for March 2026 reinforce the dominant status of the USD. With 51.14% market share of total payment transactions and 58.5% of all trade finance transactions, the USD maintains its dominance as the principal reserve currency and settlement currency in the foreseeable future. The US represents 15.59% of the SWIFT payment orders in the country breakdown. Such dominance of the US enables it to exercise strong political clout and impose sanctions (such as those against Russia in 2022), while at the same time encouraging other nations to seek alternatives.

C.2 CNY/RMB in Global Payments and Trade Finance (Q10)

RMB Metric	Value	Context
Global Payment Share	3.10%	Rank 5 globally; gradual but consistent growth
Trade Finance Share	8.04%	Rank 4; stronger in trade than general payments
Offshore RMB (Economy)	75.23% via HK	Dominated by Hong Kong hub
FX Spot (RMB)	39.47%	Significant in FX spot transactions
Avg Global Payment Share (All Reports)	2.93%	Range: 2.73%–3.13%; stable consolidation

Table 17: RMB/CNY Role in Global Payments – Summary (P2-Q10)

The higher percentage of trade financing (8.04%) than payments (3.10%) for the CNY shows the planned efforts by China to internationalize the RMB via commodity trade agreements with countries that are exporting. Although the pace of the process has been slow, the continued presence in the Top-5 in payments share and high percentage of trade financing show that the process is gaining traction.

C.3 Sanctions, BRICS Pay, and Currency Markets (Q11)

The issues of sanctions, alternative settlement systems, and BRICS Pay deliberations are strategically critical for currency markets analysis since they threaten the institutional framework behind the dominance of the USD. After Russia got disconnected from SWIFT in 2022, it highlighted the strength and vulnerability of the USD-based infrastructure of international payments – it spurred on the efforts of Russia, China, India, among other countries, to create an alternative system of payments. The key implications are as follows:

- A gradual move to BRICS Pay or RMB-based settlement would mean a reduction in the global demand for the USD; it would have an adverse effect on its exchange rate.
- The countries possessing USD reserves are exposed to "sanctions risks" since their USD reserves can be blocked anytime. It forces countries to diversify and hold more gold and other currencies.
- Since USD accounts for 51.14% of global payments (Table 8, Q19), even a 5-10% decrease in its position over a decade will be a structural change.

C.4 Offshore RMB and Hong Kong's Role (Q12)

Offshore RMB transactions data in March 2026 shows that the percentage contribution made by Hong Kong to all off-shore RMB payments is 75.57%, which is much higher than that made by United Kingdom (approximately 7%) and Singapore (approximately 4%). The reason for this high level of concentration is that Hong Kong occupies an exceptional position as China's international financial window with significant liquidity in RMB instruments, a large number of institutional investors from China, and a high degree of connectedness with China's capital markets through Stock Connect and Bond Connect programs. Although being an important advantage, this high level of concentration is also a point of concentration risk.

Section D: Business Interpretation and Recommendation

D.1 Most Event-Sensitive Market (Q13)

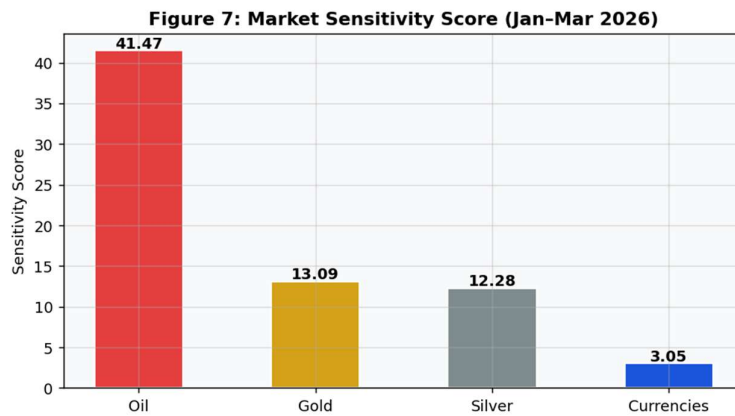


Figure 7: Market Sensitivity Score – Jan to Mar 2026

Rank	Market	Monthly Return Volatility (%)	Max Monthly Change (%)	Sensitivity Score
1st (Most Sensitive)	Brent Crude Oil	34.43%	52.03%	41.47
2nd	Gold	14.14%	11.51%	13.09
3rd	Silver	8.59%	17.81%	12.28
4th (Least Sensitive)	Currencies (CNY)	1.61%	5.21%	3.05

Table 18: Market Sensitivity Analysis (Jan–Mar 2026) (P2-Q13)

Considering the quantitative sensitivity analysis carried out during January–March 2026, it has been concluded that Brent Crude Oil is definitely the market which is more sensitive towards events around the world. The monthly volatility of Brent Crude Oil is 34.43% and the maximum change that can take place in one month is 52.03%. These figures are way higher compared to the returns of gold (14.14% and 11.51%) and the currencies (1.61% and 5.21%).

D.2 Risk Note for Companies Exposed to Oil Prices (Q14)

Risk Note: Oil Price Exposure
Oil-dependent companies such as logistics firms, airlines, manufacturing, and food distributors are exposed to the fluctuations in the Brent Oil price immediately. The current oil price at \$102.86 per barrel represents a 64% rise from the historical average due to the geopolitical factors associated with the Iran-Israel conflict and the Hormuz Strait situation. Industries most vulnerable to fluctuations in oil price include: ● Logistics and shipping. ● Airlines. ● Chemicals and plastic manufacturing industries. ● Food producing industries, including fertilizers manufacturing and transportation. Risk Management Recommendations: ● Regular monitoring of Brent Oil prices once a month and setting up of internal alerts (for example, \$110 per barrel). ● Creating a financial reserve capable of covering 3-6 months of increased expenses on fuel. ● Creation of flexible pricing terms in supplier and customer contracts allowing for cost pass-through during price spikes.

D.3 Risk Note for Companies Holding USD, Gold, or RMB/CNY Assets (Q15)

Asset Class	Key Risks	Short-Term Outlook	Long-Term Outlook
USD	Trump-era trade wars; de-dollarisation pressure; BRICS Pay alternatives	Stable – still dominant reserve currency	Monitor carefully; gradual erosion possible
Gold	Price at historic high; short-term correction risk	Elevated – geopolitical support	Strong long-term inflation hedge; recommended for reserves
RMB/CNY	Trade war with US; strong CNY reduces export competitiveness; capital controls	Limited growth in international role	Useful for diversification, not primary reserve

Table 19: Asset Risk Assessment – USD, Gold, and RMB/CNY (P2-Q15)

Conclusion: USD is the most suitable in the short term because of its liquidity and wide acceptance around the world. Gold is the most suitable in the long term since it will act as a hedge against inflation and economic crises. RMB/CNY provides diversification benefits, though it poses some risks.

5. Final Recommendations

Considering the comprehensive analysis of the data and market research results found in this report, the following three suggestions are made for the investors, firms, and government officials respectively.

Recommendation 1 – For Investors: Diversify with a Gold-Anchored Strategy

The numbers speak for themselves when it comes to gold, which has outpaced all other asset classes examined in this report from a long-term perspective, with its current value (4,712.89/oz) being 1,085% higher than the historical long-term average. In spite of this, the underlying factors supporting gold, including geopolitical risks, de-dollarisation concerns, and the accumulation of gold by central banks, continue to persist. One should:

- Maintain significant allocations to gold (10%-20%) as an inflation hedge and crisis insurance.
- Diversify holdings among US stocks, gold, and silver (10%-20%), benefiting from the safe haven and green energy industrial demand for the latter.
- Avoid overweight positions in USD-denominated securities due to long-term risks of de-dollarisation while giving preference to USD holdings on the short end.
- Regard RMB/CNY as a diversification currency and not a reserve one until its international payment share grows beyond 3%.

Recommendation 2 – For Companies: Proactive Oil Risk Management and Supply Chain Resilience

Given that Brent Oil is priced at \$102.86/barrel — well above its historic average and highly volatile due to geopolitical factors (sensitivity rating of 41.47) — companies with an exposure to the oil market need to view price risk strategically, at the board level. Recommendations:

- Focus on oil price monitoring using pre-determined responses at various price points (\$100, \$120, \$150/barrel).
- Prepare for contingencies by creating a financial buffer of up to 3-6 months of additional costs from higher-than-budgeted oil prices.
- Make energy efficiency investments that lower the reliance on oil in the long term, both as a strategy and a competitive advantage.
- Conclude energy procurement agreements that leave some flexibility in pricing while oil prices are high.
- Companies with logistical or manufacturing activities should focus on diversification of their activities geographically to minimize dependence on one route that could be blocked via the Hormuz Strait.

Recommendation 3 – For Policymakers: Monitor the De-dollarisation Trend and Strengthen Domestic Financial Infrastructure

While USD dominance persists with a 51.14% payment share in SWIFT statistics, the constant inclusion of the CNY among the Top-5 currencies and the emergence of an offshore RMB payment system, where Hong Kong leads with 75.57% offshore share, indicate a steady yet noticeable shift in the international monetary system. Policy-makers within central banks, trade ministries, and financial regulatory bodies should:

- Monitor monthly changes in SWIFT payment share data for signs of RMB internationalisation or USD dominance erosion.
- Consider partial gold reserve buildup as a measure to reduce exposure to any possible future sanctions danger, taking into account the established practice of China, Russia, India, and Turkey.
- Build financial infrastructure capable of supporting payments in several currency systems to ensure resilience no matter which payment system will dominate in the coming decade.
- Conduct stable domestic monetary and fiscal policy to establish international investor confidence – the key element of currency trust as evident from the constant 21.3% global payment share of EUR.

6. Appendix – Bonus Analytics

B1 & B2: Correlation Analysis of Annual Average Prices

The annual average prices of Gold, Silver, and Brent Oil were determined, and a correlation analysis using Pearson Correlation was done.

Asset Pair	Correlation Coefficient (r)	Interpretation
Gold vs. Silver	0.959	Very strong positive correlation – move together closely
Gold vs. Brent Oil	0.458	Moderate positive correlation – some co-movement
Silver vs. Brent Oil	0.515	Moderate positive correlation – some co-movement

Table B1: Correlation Matrix – Annual Average Prices (Bonus Q2)

The extraordinarily high correlation coefficient of Gold-Silver (0.959) proves beyond any shadow of doubt that gold and silver move together as a well-coupled pair. The moderate correlation between oil and precious metals (0.458-0.515) indicates partial co-movement – presumably because of inflation expectations – but also periods of divergence (such as the COVID-19 period where oil fell while gold rose).

B3: Correlation Heatmap

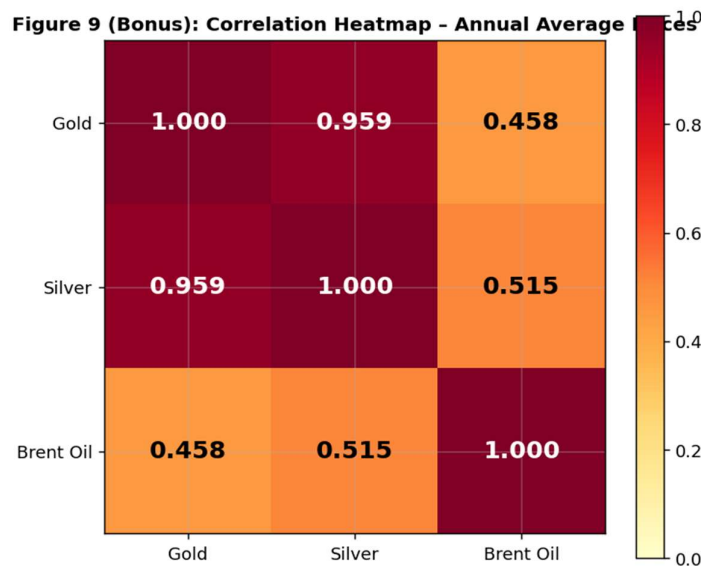


Figure 9: Correlation Heatmap – Annual Average Prices (Bonus Q3)

B6: Simple Price Forecast (Trending Estimate)

Linear trend analysis forecasting was done for both the annual average price of Brent Oil and Gold. These are hypothetical forecasts based solely on linear trends in the past, and SHOULD NOT be considered surefire forecasts. There are unforeseeable occurrences in the markets that could not be predicted using linear projection.

Year	Brent Oil Estimate (\$/barrel)	Gold Estimate (\$/troy oz)
2027	\$98.49	\$1,315.28
2028	\$99.36	\$1,331.05
2029	\$100.23	\$1,346.82
2030	\$101.09	\$1,362.59
2031	\$101.96	\$1,378.36

Table B2: Simple Trendline Forecast – Brent Oil and Gold (Bonus Q6) – Illustrative Only

Please note that these predictions have been made using historical trends and do not include any information about current prices, which are much higher than the trend line. The actual future prices will be influenced by geopolitics, OPEC decisions, monetary policies, and many other factors not included in the linear model.

7. Conclusion

These reports analyse the datasets of the following: Brent Crude Oil (from 1946 until 2026); Gold and Silver (from 1915 until 2026); SWIFT global currency payments (in 2026).

The main analysis finding is that commodity markets are very sensitive to geopolitical events. For example, the sensitivity of Oil is 41.47, while the sensitivity of Gold is 13.09 and of Currencies is 3.05. But the long-term structural trend seems even more relevant. In particular, the premium of Gold over its historical mean is 1,085%, Silver reached the all-time highs, and the process of internationalisation of Chinese Renminbi shows that the world is changing from pure USD-centred finance to multipolar one.

The messages for non-technical managers are as follows:

- Oil is expensive and risky – hedge it and/or keep money in reserve.
- Gold is the best proven safe haven – keep it in portfolios and reserves.
- The USD reign will be over soon, but others develop – learn and diversify.

The connection between these three financial markets highlights an important fact of finance today: The price of commodities, currencies, and geopolitics is one story told from three angles. Those analysts and businesspeople who grasp these connections will be in the best position to face the future.

8. References

- Brent Oil Dataset: Historical monthly Brent Crude Oil prices, 1946–2026. Provided by course instructor.
- Gold 100 Years Dataset: Historical monthly Gold prices (USD/troy oz), 1915–2026. Provided by course instructor.
- Silver 100 Years Dataset: Historical monthly Silver prices (USD/troy oz), 1915–2026. Provided by course instructor.
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